Act2CJMarcelaGuerrero

October 19, 2026

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#LECCIÓN 2 : CLUSTERING JERÁRQUICO
#MARCELA GUERRERO HERNÁNDEZ
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[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os
```

[54]:
```
data0 = pd.read_excel('Amazon.xlsx')
data0.head()

# Guardar nombres y datos numéricos por separado
nombres_clientes = data0['Unnamed: 0']  # Guardar nombres
data = data0.drop('Unnamed: 0', axis=1)  # Quitar columna de nombres
data.head()
```

[54]:

	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Calidad Producto	Numero Estrellas
0	205	3	345	235	24	23	26	21	17
1	9	15	315	33	25	4	42	215	28
2	17	26	285	3	43	27	41	26	33
3	135	5	355	295	18	23	39	195	17
4	3	45	48	39	34	46	225	34	43

[10]:
```
from sklearn.preprocessing import normalize
data_scaled = normalize(data)
data_scaled
```

[10]:
```
array([[0.43826336, 0.00641361, 0.73756517, 0.50239946, 0.05130888,
        0.04917101, 0.05558462, 0.04489527, 0.03634379],
       [0.02323527, 0.03872544, 0.81323434, 0.08519598, 0.06454241,
```

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```

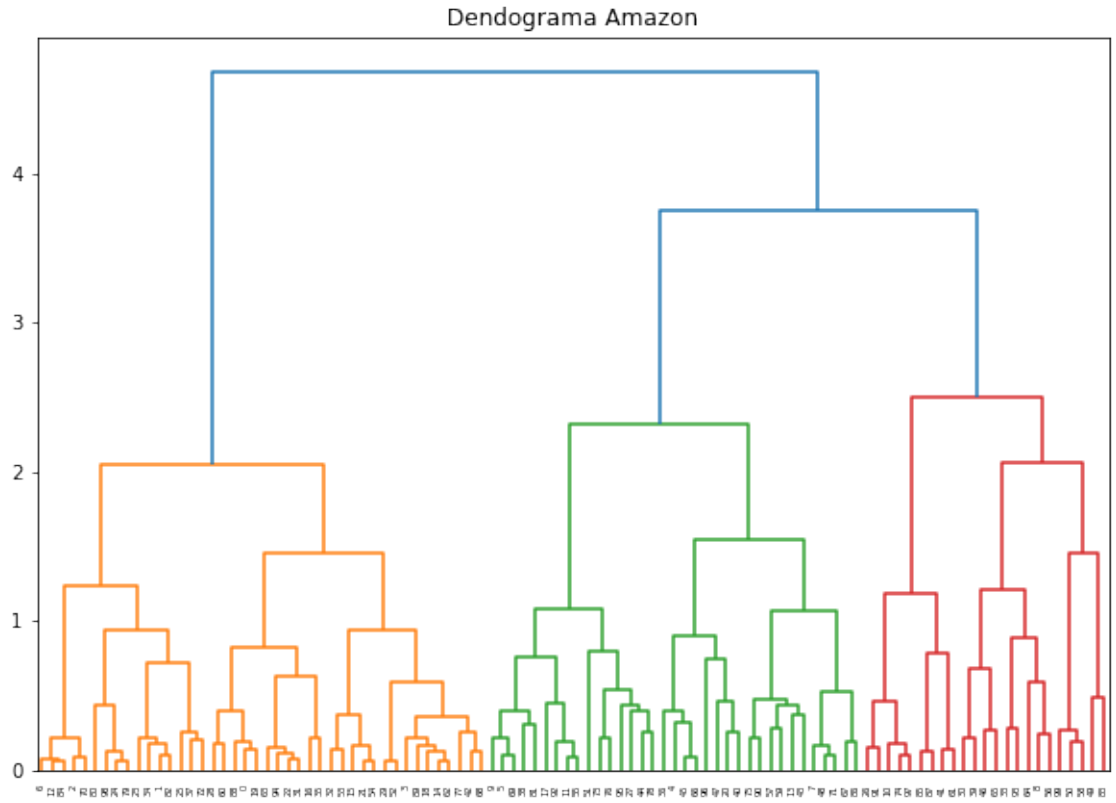
[11]:
```
data_scaled = pd.DataFrame(data_scaled, columns = data.columns)
data_scaled.head()
```

[11]:

	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	\
0	0.438263	0.006414	0.737565	0.502399	0.051309	
1	0.023235	0.038725	0.813234	0.085196	0.064542	
2	0.057235	0.087535	0.959520	0.010100	0.144770	
3	0.258856	0.009587	0.680696	0.565649	0.034514	
4	0.011975	0.179625	0.191600	0.155675	0.135717	

	Servicio Retorno	Tamano Paquete	Calidad Producto	Numero Estrellas
0	0.049171	0.055585	0.044895	0.036344
1	0.010327	0.108431	0.555065	0.072287
2	0.090902	0.138036	0.087535	0.111102
3	0.044101	0.074781	0.373904	0.032597
4	0.183617	0.898127	0.135717	0.171642

[13]:
```
import scipy.cluster.hierarchy as shc
plt.figure(figsize=(10,7))
plt.title("Dendograma Amazon")
dend = shc.dendrogram(shc.linkage(data_scaled, method= 'ward'))
```



[14]:
```
print(dend['color_list'])
```

```
['C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1', 'C1',
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'C3', 'C3', 'C3', 'C3', 'C3', 'C3', 'C0', 'C0']
```

[19]:
```
colores_unicos = set(dend['color_list'])
colores_unicos
```

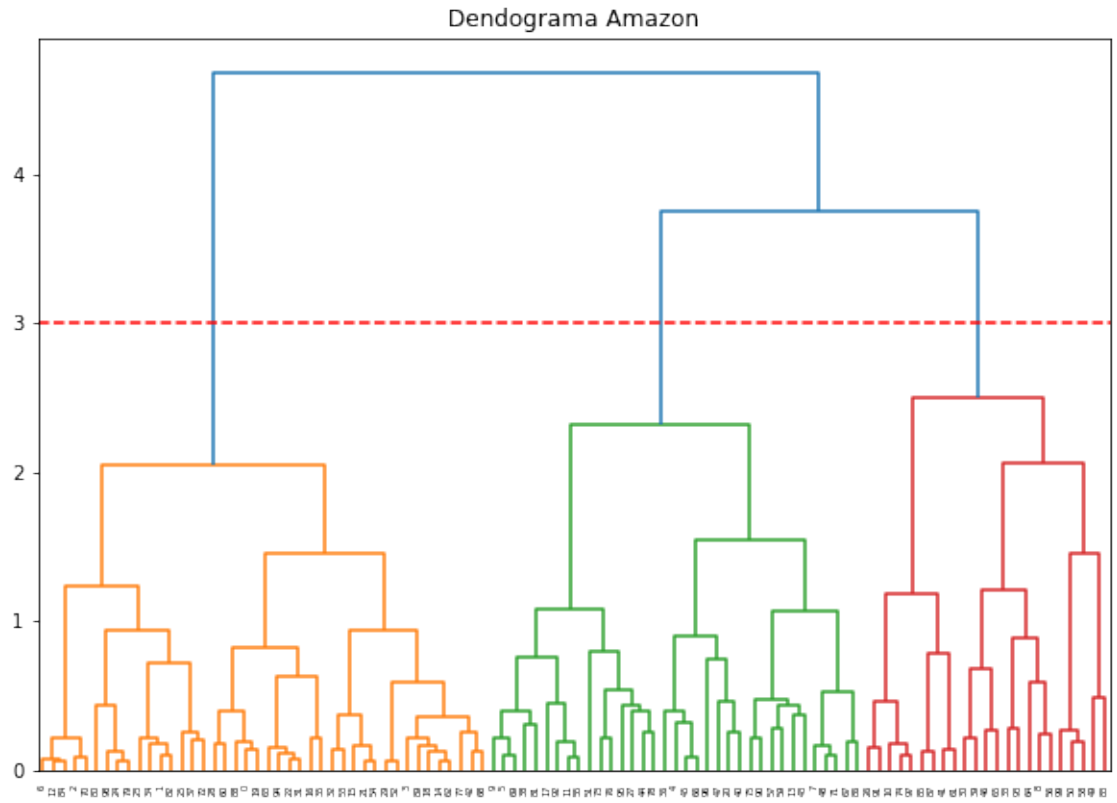
[19]:
```
{'C0', 'C1', 'C2', 'C3'}
```

[20]:
```
num_clusters_optimo = len(colores_unicos)-1
num_clusters_optimo
```

[20]:
```
3
```

[28]:
```
plt.figure(figsize=(10,7))
plt.title("Dendograma Amazon")
dend = shc.dendrogram(shc.linkage(data_scaled, method = 'ward'))
plt.axhline(y=3, color = 'r', linestyle = '--')
```

[28]:
```
<matplotlib.lines.Line2D at 0x7d89b61a30a0>
```



[23]:
```
data
```

[23]:

	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	\
0	205	3	345	235	24	
1	9	15	315	33	25	
2	17	26	285	3	43	
3	135	5	355	295	18	
4	3	45	48	39	34	
..	…	…	…	…	…	
95	3	8	32	25	7	
96	305	25	46	24	33	
97	1	14	26	25	24	
98	155	11	335	34	26	
99	125	9	45	25	22	

	Servicio Retorno	Tamano Paquete	Calidad Producto	Numero Estrellas
0	23	26	21	17
1	4	42	215	28
2	27	41	26	33
3	23	39	195	17
4	46	225	34	43
..	…	…	…	…
95	21	42	17	1
96	28	355	26	45
97	27	42	185	23
98	29	42	215	27
99	3	3	22	18

[100 rows x 9 columns]

[30]:
```
from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters = 3, affinity='euclidean', linkage␣
 ↪= 'ward')
grupos = cluster.fit_predict(data_scaled)
grupos
```

[30]:
```
array([1, 1, 1, 1, 2, 2, 1, 2, 0, 2, 0, 2, 1, 2, 1, 1, 1, 2, 1, 1, 2, 1,
       1, 1, 1, 1, 0, 2, 1, 1, 0, 1, 1, 0, 1, 1, 2, 1, 2, 0, 2, 0, 1, 2,
       2, 2, 0, 2, 2, 0, 0, 2, 1, 1, 1, 2, 0, 2, 0, 2, 1, 0, 1, 1, 0, 0,
       2, 2, 1, 2, 1, 2, 1, 2, 0, 2, 2, 1, 2, 1, 1, 2, 1, 0, 1, 0, 2, 0,
       1, 1, 2, 0, 2, 0, 1, 2, 2, 0, 1, 0])
```

[34]:
```
#SEPARACIÓN DE GRUPOS MEDIANTE PCA
campos = data_scaled.values
print(campos)
```

```
[[0.43826336 0.00641361 0.73756517 0.50239946 0.05130888 0.04917101
  0.05558462 0.04489527 0.03634379]
 [0.02323527 0.03872544 0.81323434 0.08519598 0.06454241 0.01032679
  0.10843125 0.55506471 0.0722875 ]
 [0.05723452 0.08753514 0.95951982 0.01010021 0.14476966 0.09090188
  0.13803618 0.08753514 0.11110229]
 [0.25885639 0.00958727 0.68069644 0.56564916 0.03451419 0.04410146
  0.07478074 0.37390368 0.03259673]
 [0.01197502 0.17962533 0.19160036 0.15567529 0.13571692 0.18361701
  0.89812667 0.13571692 0.17164198]
 [0.14494006 0.251738   0.60264551 0.03661644 0.03966781 0.02898801
  0.73995715 0.03356507 0.00457705]
 [0.04755854 0.02481315 0.98218723 0.06823617 0.07237169 0.09304932
  0.07857498 0.05996511 0.06410064]
 [0.13323393 0.04304481 0.06354234 0.52268696 0.05739308 0.04509456
```

```
 0.70716471 0.44069685 0.05944283]
[0.73804829 0.0214705  0.12613916 0.63069581 0.09393342 0.00805144
 0.10198485 0.07246292 0.12882297]
[0.00348086 0.30457545 0.56564012 0.00522129 0.06439595 0.0556938
 0.75708755 0.04699164 0.06787681]
[0.05320841 0.03547227 0.19509749 0.10641681 0.00886807 0.12415295
 0.12858698 0.9533173  0.07537858]
[0.30081705 0.01696917 0.70190646 0.03548099 0.00462795 0.03856629
 0.6402004  0.03856629 0.04936485]
[0.03421592 0.01710796 0.98981777 0.0464359  0.05132388 0.03421592
 0.08065182 0.05376788 0.05865587]
[0.37889397 0.15360566 0.08806725 0.58370152 0.05529804 0.07577879
 0.68610529 0.05120189 0.04710574]
[0.33000964 0.09127926 0.69512668 0.47043927 0.00421289 0.0365117
 0.04774608 0.41426742 0.05476756]
[0.02878814 0.00169342 0.8213087  0.3979537  0.04572234 0.02878814
 0.04064208 0.3979537  0.05757628]
[0.03645394 0.46706613 0.64933583 0.58098469 0.08202137 0.06607277
 0.07062951 0.05012417 0.05240254]
[0.41749646 0.0153366  0.65606587 0.36637445 0.05793828 0.02556101
 0.50269982 0.04771388 0.04260168]
[0.3782489  0.00999148 0.69226685 0.43534307 0.04710269 0.05566682
 0.04853005 0.42106953 0.05566682]
[0.36323349 0.10046884 0.76510885 0.51780094 0.00463702 0.04018754
 0.05255293 0.00463702 0.0061827 ]
[0.386348   0.10536764 0.10068463 0.00468301 0.04917156 0.04214705
 0.73757346 0.52683818 0.06087908]
[0.03653752 0.00429853 0.89194526 0.26865821 0.02579119 0.03653752
 0.05588091 0.35462884 0.00429853]
[0.02579636 0.00343951 0.78248953 0.61051381 0.0601915  0.05847175
 0.0722298  0.04471369 0.00687903]
[0.03042923 0.19018267 0.84948258 0.06085845 0.04817961 0.06339422
 0.09128768 0.46911724 0.05325115]
[0.45206655 0.01240967 0.77117236 0.04254744 0.05850273 0.04609306
 0.03368339 0.43433845 0.06027554]
[0.04106056 0.18745039 0.70517053 0.05177201 0.06069822 0.04998677
 0.41953184 0.52664635 0.06069822]
[0.05696103 0.35600642 0.15664282 0.11392205 0.09018829 0.11866881
 0.17088308 0.87814917 0.0996818 ]
[0.03874633 0.09686582 0.72276808 0.45452426 0.04768779 0.05811949
 0.49923156 0.04321706 0.05811949]
[0.31413959 0.02513117 0.88856626 0.31413959 0.05564758 0.03051642
 0.04846725 0.04846725 0.06103283]
[0.36927647 0.3332495  0.53139785 0.49537087 0.0702526  0.00540405
 0.07565664 0.4593439  0.05584181]
[0.04750309 0.05066997 0.00950062 0.83922133 0.09817306 0.00950062
 0.01266749 0.52253404 0.08867244]
[0.02461563 0.03340693 0.78242552 0.60659956 0.05802257 0.05626431
```

```
 0.07208864 0.04395649 0.06681386]
[0.04103525 0.00157828 0.73389965 0.46559225 0.05839632 0.03787869
 0.03630041 0.48137504 0.07102255]
[0.04861252 0.52901858 0.09150592 0.81497458 0.1000846  0.09722504
 0.12010152 0.05433164 0.09150592]
[0.02738021 0.01140842 0.87844842 0.03878863 0.03878863 0.02509853
 0.07073221 0.46774526 0.00456337]
[0.01909709 0.35011336 0.79571219 0.47742731 0.05304748 0.05092558
 0.08063217 0.03819419 0.05092558]
[0.03823337 0.00424815 0.0615982  0.0615982  0.07859081 0.0531019
 0.98769534 0.05097782 0.0615982 ]
[0.00312254 0.07025712 0.71037754 0.04215427 0.03747046 0.040593
 0.5698633  0.39812367 0.04839935]
[0.         0.17312925 0.56885325 0.04451895 0.01813735 0.0428701
 0.73373825 0.32152575 0.0230839 ]
[0.0418933  0.00349111 0.11171547 0.78549943 0.07331328 0.07680439
 0.15360878 0.57603291 0.04538441]
[0.21476784 0.03843214 0.08590714 0.05199642 0.05877857 0.05651785
 0.87037493 0.41823211 0.05651785]
[0.57760365 0.08810903 0.09398297 0.07636116 0.06657127 0.09006701
 0.44054515 0.65592278 0.08419307]
[0.37318997 0.1751708  0.70829934 0.34272549 0.05483608 0.01980192
 0.04721996 0.44935119 0.05788253]
[0.07929518 0.20616748 0.13638772 0.74537474 0.09832603 0.07929518
 0.58678437 0.07612338 0.10466964]
[0.00176146 0.022899   0.57247507 0.32587042 0.04227508 0.02994485
 0.74862124 0.02818339 0.04051362]
[0.07177865 0.35889325 0.13494386 0.06603636 0.1062324  0.04019604
 0.90441099 0.00861344 0.11197469]
[0.40592889 0.24879512 0.01309448 0.58925161 0.0680913  0.08380467
 0.04975902 0.64162953 0.01047558]
[0.03079523 0.35323942 0.05072156 0.05072156 0.06521343 0.04166414
 0.82422532 0.42569879 0.05072156]
[0.06073355 0.00209426 0.09214746 0.47120858 0.00628278 0.05026225
 0.70157722 0.51309379 0.08795893]
[0.21268415 0.82710504 0.03150876 0.11815786 0.29933325 0.11028067
 0.20480696 0.14966663 0.29933325]
[0.9228913  0.17460106 0.20453267 0.0149658  0.10476063 0.12471504
 0.12970364 0.12471504 0.12970364]
[0.13614992 0.25135369 0.42939589 0.26182676 0.37703053 0.26182676
 0.47128817 0.27229983 0.39797667]
[0.37464062 0.34133923 0.52449687 0.49119548 0.07159799 0.05661236
 0.07326305 0.45789409 0.05827743]
[0.03051268 0.02615372 0.73012479 0.53397186 0.05448692 0.0566664
 0.10025594 0.40320324 0.03705111]
[0.03823712 0.00804992 0.87542873 0.29180958 0.03219968 0.04226208
 0.05634944 0.37230877 0.04829952]
[0.25536481 0.02289478 0.67803761 0.06163978 0.04931183 0.06340092
```

```
 0.67803761 0.03698387 0.05635637]
[0.56428699 0.05067067 0.08521885 0.79460821 0.10594776 0.00921285
 0.11055419 0.07139958 0.10825097]
[0.05482112 0.25380148 0.09745977 0.55836326 0.00812165 0.00609124
 0.78170857 0.00609124 0.01015206]
[0.94018329 0.03935651 0.16617193 0.11806953 0.13556131 0.10932364
 0.0962048  0.12244248 0.13556131]
[0.20596919 0.40298321 0.00716415 0.42089357 0.0591042  0.0394028
 0.77910087 0.04477591 0.06268628]
[0.26868493 0.16467786 0.85805831 0.39002651 0.04506973 0.05373699
 0.03293557 0.04160283 0.06760459]
[0.53817713 0.20049736 0.09708293 0.06120446 0.07597795 0.04854147
 0.47486217 0.64370206 0.09497243]
[0.3198163  0.08580437 0.72543696 0.42902186 0.03900199 0.04212215
 0.05772294 0.41342106 0.04992254]
[0.04023907 0.05096949 0.73771626 0.65723812 0.09120856 0.06974772
 0.00804781 0.0563347  0.0563347 ]
[0.22130333 0.0040237  0.14485309 0.94556878 0.06437915 0.1287583
 0.02011848 0.06840285 0.10059242]
[0.47875596 0.01811509 0.11645415 0.58227076 0.06728462 0.05952101
 0.08798758 0.63402816 0.00776361]
[0.05301222 0.31554893 0.11612201 0.07825613 0.08330492 0.09845127
 0.92140288 0.00757317 0.11107322]
[0.01460148 0.41066668 0.05840593 0.48367409 0.00547556 0.04562963
 0.64794077 0.41066668 0.05658074]
[0.45278576 0.14523317 0.72616585 0.31609572 0.05980189 0.03246388
 0.04100701 0.36735449 0.0734709 ]
[0.17544559 0.28223856 0.63312973 0.03966596 0.00457684 0.03508912
 0.69415429 0.03661473 0.05187087]
[0.05820916 0.08731374 0.95398343 0.10024911 0.1455229  0.09378142
 0.13582137 0.08731374 0.11318448]
[0.07616433 0.0299217  0.11152634 0.42162396 0.01088062 0.04352247
 0.72084096 0.53043014 0.01088062]
[0.0296482  0.01811835 0.81532558 0.03953094 0.04776655 0.03129533
 0.40354498 0.40354498 0.05929641]
[0.03814285 0.09535713 0.66749994 0.33008239 0.04841208 0.03960989
 0.53546698 0.37409337 0.06601648]
[0.06840051 0.00456003 0.15048113 0.15048113 0.10944082 0.12312092
 0.1869614  0.934807   0.11856089]
[0.04225181 0.02414389 0.09456358 0.49293779 0.06438371 0.05432376
 0.85509617 0.05231177 0.06840769]
[0.0299383  0.0063028  0.6539156  0.48058857 0.03466541 0.04096821
 0.41756056 0.40180356 0.04254391]
[0.27590777 0.02173819 0.81100164 0.27590777 0.04849288 0.02508252
 0.04347638 0.42640292 0.05350939]
[0.00737441 0.14011373 0.52358287 0.3318483  0.02212322 0.04572132
 0.73006626 0.24335542 0.03539715]
[0.40430304 0.01437522 0.78165255 0.04132876 0.05570397 0.03773495
```

```
 0.06109468 0.45821011 0.073673  ]
[0.44633507 0.01460733 0.70602093 0.0308377  0.05842932 0.03408377
 0.39764397 0.36518324 0.07141361]
[0.03994652 0.05404529 0.64619364 0.0963416  0.00939918 0.10339098
 0.74018544 0.06579426 0.07519344]
[0.02133038 0.03732816 0.81322057 0.0853215  0.06132483 0.10131928
 0.10931818 0.54659088 0.06932372]
[0.50194895 0.8074831  0.16586139 0.10911934 0.01309432 0.10911934
 0.16149662 0.09602502 0.09602502]
[0.03014465 0.03478229 0.98549814 0.00695646 0.06492694 0.06492694
 0.07883985 0.06492694 0.08811513]
[0.44300067 0.54932083 0.12404019 0.07442411 0.09923215 0.07796812
 0.15948024 0.65564099 0.09923215]
[0.02188841 0.26448495 0.07660943 0.53809007 0.04924892 0.04924892
 0.61105144 0.50160939 0.06566523]
[0.34454946 0.5742491  0.12141267 0.07875416 0.09187986 0.07547274
 0.11813124 0.70550604 0.06890989]
[0.29097875 0.0120405  0.73246376 0.61205876 0.0040135  0.05016875
 0.008027   0.0040135  0.03812825]
[0.31199059 0.18138988 0.67477034 0.45710249 0.04933805 0.00580448
 0.0536914  0.4425913  0.06530036]
[0.02799536 0.026129   0.07278793 0.66255678 0.00559907 0.07092157
 0.73721106 0.04105986 0.06345614]
[0.08089801 0.28651378 0.12808851 0.07078576 0.11123476 0.0471905
 0.09775176 0.92695635 0.08089801]
[0.36452026 0.04938662 0.59969463 0.091718   0.08466277 0.00940697
 0.69376437 0.06114533 0.06584882]
[0.31413757 0.44640602 0.08266778 0.81014426 0.07274765 0.08266778
 0.13557516 0.0595208  0.06944094]
[0.00491935 0.06149192 0.82399169 0.55342725 0.05411289 0.05165321
 0.06149192 0.00491935 0.03935483]
[0.04593152 0.12248406 0.48993623 0.38276268 0.10717355 0.32152065
 0.6430413  0.26027862 0.01531051]
[0.64024082 0.05247876 0.09656091 0.05037961 0.06927196 0.05877621
 0.74519833 0.05457791 0.09446176]
[0.00504299 0.0706018  0.13111762 0.12607464 0.12103165 0.13616061
 0.21180539 0.93295232 0.11598867]
[0.35770734 0.02538568 0.7731094  0.07846483 0.06000252 0.06692589
 0.09692715 0.49617469 0.06231031]
[0.89135765 0.06417775 0.32088875 0.17827153 0.15687895 0.02139258
 0.02139258 0.15687895 0.1283555 ]]
```

[73]:
```
from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(campos)
campos = pca.transform(campos)
campos
```

```
[73]: array([[-0.39993219,  0.06035374],
             [-0.40829583, -0.06402268],
             [-0.49703182, -0.35955286],
             [-0.33996675,  0.18434501],
             [ 0.58207751, -0.38797653],
             [ 0.1723403 , -0.52962573],
             [-0.56528441, -0.33770585],
             [ 0.5335887 ,  0.06994712],
             [ 0.09981604,  0.47165743],
             [ 0.22195207, -0.56503313],
             [ 0.08274005,  0.39712712],
             [-0.00143465, -0.48936583],
             [-0.57292039, -0.35927774],
             [ 0.51350639,  0.01301312],
             [-0.35272162,  0.20639728],
             [-0.4688626 ,  0.0168001 ],
             [-0.25142488,  0.04599019],
             [-0.05456876, -0.22302796],
             [-0.36123131,  0.20553368],
             [-0.41087067,  0.02173183],
             [ 0.54207892, -0.03308916],
             [-0.5138909 , -0.0900875 ],
             [-0.4284552 , -0.0601418 ],
             [-0.42320562, -0.09945498],
             [-0.42610215,  0.06318626],
             [-0.10615885, -0.20376697],
             [ 0.19058214,  0.3966612 ],
             [-0.096838  , -0.33104612],
             [-0.51505714, -0.10893109],
             [-0.17050111,  0.33261099],
             [ 0.13793391,  0.63651115],
             [-0.42206647, -0.05789717],
             [-0.40326074,  0.12516351],
             [ 0.21346687,  0.37300285],
             [-0.48923475, -0.13003385],
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             [-0.03811566, -0.37986053],
             [ 0.18646361, -0.43630007],
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             [ 0.36295672,  0.28878811],
             [-0.34697878,  0.19244643],
             [ 0.42217536,  0.03448406],
             [ 0.16141571, -0.48420739],
             [ 0.65108758, -0.4197626 ],
             [ 0.19657428,  0.70259997],
```

```
[ 0.66940341, -0.17418122],
[ 0.50314121,  0.04478847],
[ 0.37827002,  0.22768866],
[ 0.09257978,  0.29000069],
[ 0.15947025, -0.08591885],
[-0.16414836,  0.33801587],
[-0.35892963,  0.07853561],
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[ 0.04236318, -0.50393996],
[ 0.13801031,  0.49375593],
[ 0.57613865, -0.17095284],
[ 0.0783886 ,  0.35710609],
[ 0.67312488, -0.09618175],
[-0.47969491, -0.06068297],
[ 0.39939783,  0.25539679],
[-0.3695398 ,  0.16814344],
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[ 0.02636687,  0.46187498],
[ 0.10479979,  0.62527833],
[ 0.66976445, -0.42777742],
[ 0.55740005,  0.0791324 ],
[-0.36929226,  0.16505085],
[ 0.12646304, -0.50126007],
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[-0.23024785, -0.32414928],
[-0.02497426, -0.23000985],
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[ 0.59247292, -0.24149892],
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[ 0.2099264 , -0.34219382],
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[-0.14855054, -0.1595983 ],
[ 0.11014101, -0.56579396],
[-0.40616865, -0.06841998],
[ 0.23672835,  0.24043862],
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[ 0.23730948,  0.44802811],
[ 0.49928987,  0.14100592],
[ 0.21649152,  0.46796438],
[-0.42855949,  0.06683056],
[-0.31814994,  0.22452555],
[ 0.53277044, -0.10971321],
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[ 0.11664927, -0.41801606],
[ 0.21734196,  0.44145587],
```

```
       [-0.45720098, -0.11118217],
       [ 0.18527671, -0.23439125],
       [ 0.53527538, -0.14792444],
       [ 0.19164058,  0.36824856],
       [-0.38487326,  0.03658414],
       [-0.08413049,  0.3546116 ]])
```

[37]:
```
dataframe = pd.DataFrame(grupos, columns= ['grupo'])
dataframe
```

[37]:
```
    grupo
0       1
1       1
2       1
3       1
4       2
..    ...
95      2
96      2
97      0
98      1
99      0

[100 rows x 1 columns]
```

[38]:
```
dataframe2 = pd.concat([data, dataframe], axis= 1, join = 'inner')
dataframe2
```

[38]:
```
    Velocidad Entrega  Precio  Durabilidad  Imagen Producto  Valor Educativo  \
0                 205       3          345              235               24
1                   9      15          315               33               25
2                  17      26          285                3               43
3                 135       5          355              295               18
4                   3      45           48               39               34
..                ...     ...          ...              ...              ...
95                  3       8           32               25                7
96                305      25           46               24               33
97                  1      14           26               25               24
98                155      11          335               34               26
99                125       9           45               25               22

    Servicio Retorno  Tamano Paquete  Calidad Producto  Numero Estrellas  \
0                 23              26                21                17
1                  4              42               215                28
2                 27              41                26                33
3                 23              39               195                17
4                 46             225                34                43
```

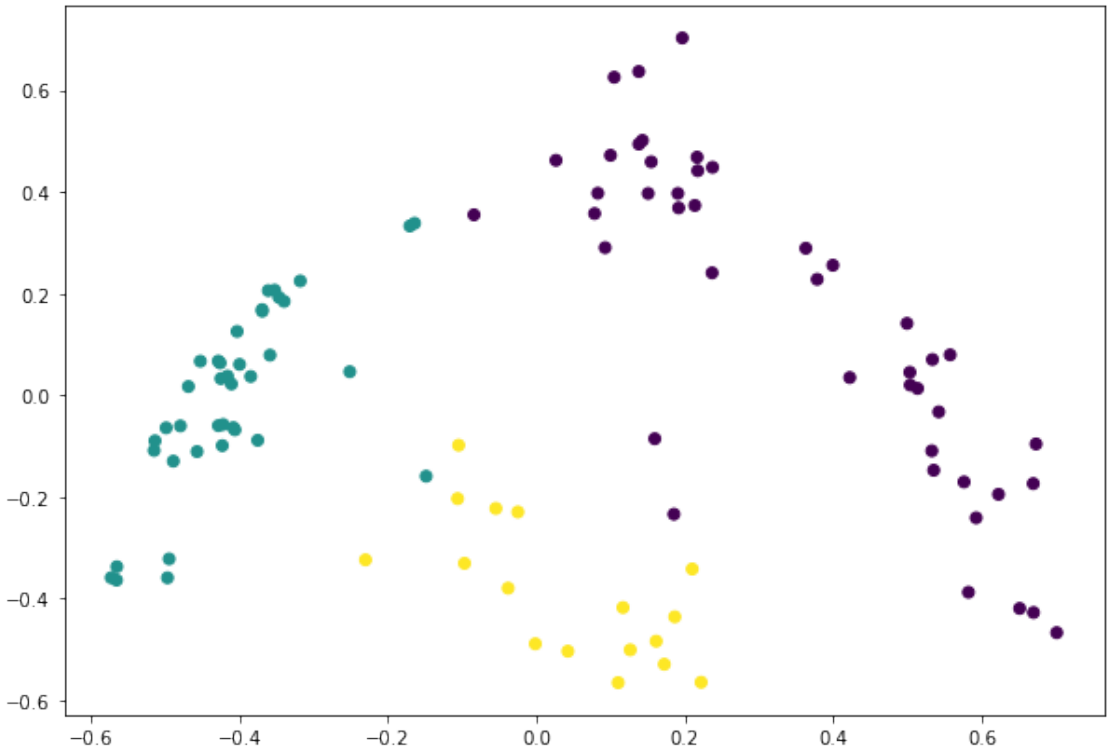
```
..                ...               ...                ...                ...
95                 21                42                 17                  1
96                 28               355                 26                 45
97                 27                42                185                 23
98                 29                42                215                 27
99                  3                 3                 22                 18

    grupo
0       1
1       1
2       1
3       1
4       2
..    ...
95      2
96      2
97      0
98      1
99      0

[100 rows x 10 columns]
```

[43]:
```
from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters = 3, affinity='euclidean', linkage␣
 ↪= 'ward')
grupos = cluster.fit_predict(dataframe2)
grupos
plt.figure(figsize = (10,7))
plt.scatter(campos[:,0], campos[:,1], c = cluster.labels_)
```

[43]:
```
<matplotlib.collections.PathCollection at 0x7d89b43e9d60>
```



[46]:
```
#REGRESAMOS LOS NOMBRES PARA SABER DE QUIÉN SE TRATA CADA OBSERVACIÓN

resultado = data.copy()
resultado.insert(0, 'Cliente', nombres_clientes.values)
resultado['grupo'] = grupos
resultado
```

[46]:

	Cliente	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	\
0	Adam	205	3	345	235	
1	Anna	9	15	315	33	
2	Bernard	17	26	285	3	
3	Edward	135	5	355	295	
4	Emilia	3	45	48	39	
..	…	…	…	…	…	
95	Teofan	3	8	32	25	
96	Teofil	305	25	46	24	
97	Teofila	1	14	26	25	
98	Teon	155	11	335	34	
99	Teresa	125	9	45	25	

	Valor Educativo	Servicio Retorno	Tamano Paquete	Calidad Producto	\
0	24	23	26	21	
1	25	4	42	215	

```
2                43                27                41                26
3                18                23                39               195
4                34                46               225                34
..              ...               ...               ...               ...
95                7                21                42                17
96               33                28               355                26
97               24                27                42               185
98               26                29                42               215
99               22                 3                 3                22

    Numero Estrellas  grupo
0                 17      1
1                 28      1
2                 33      1
3                 17      1
4                 43      0
..               ...    ...
95                 1      0
96                45      0
97                23      0
98                27      1
99                18      0

[100 rows x 11 columns]
```

[69]:
```python
#IDENTIFICAMOS A NUSTRAS CLIENTAS OBJETIVO
from scipy.spatial.distance import cdist

X = data_scaled.values
D = cdist(X, X)  # matriz de distancias 100x100

objetivos = {'Salomé': 'salome', 'Stephanía': 'stephania', 'Lydia': 'lydia'}


#PARA SABER QUÉ PRODUCTOS RECOMENDARLES, BUSCAMOS LAS OBSERVACIONES QUE ESTÉN␣
 ↪MÁS CERCA DE ELLAS DENTRO DEL CLUSTER
vecinos_dict = {}
for nombre, patron in objetivos.items():
    i = resultado.index[resultado['Cliente'].str.lower().str.
 ↪contains(patron)][0]
    g = resultado.loc[i, 'grupo']

    # candidatos: mismo cluster, sin contarse a sí misma
    mismo = resultado.index[(resultado['grupo'] == g) & (resultado.index != i)]
    cercanos = mismo[np.argsort(D[i, mismo])][:3]
    vecinos_dict[nombre] = (i, list(cercanos))
```

```
    print(f"\n{nombre} (fila {i}, cluster {g})")
    for j in cercanos:
        print(f"  -> {resultado.loc[j,'Cliente']}  (distancia = {D[i,j]:.3f})")
```

```
Salomé (fila 81, cluster 2)
  -> Magdalyna  (distancia = 0.240)
  -> Fabian  (distancia = 0.261)
  -> Kalyna  (distancia = 0.264)

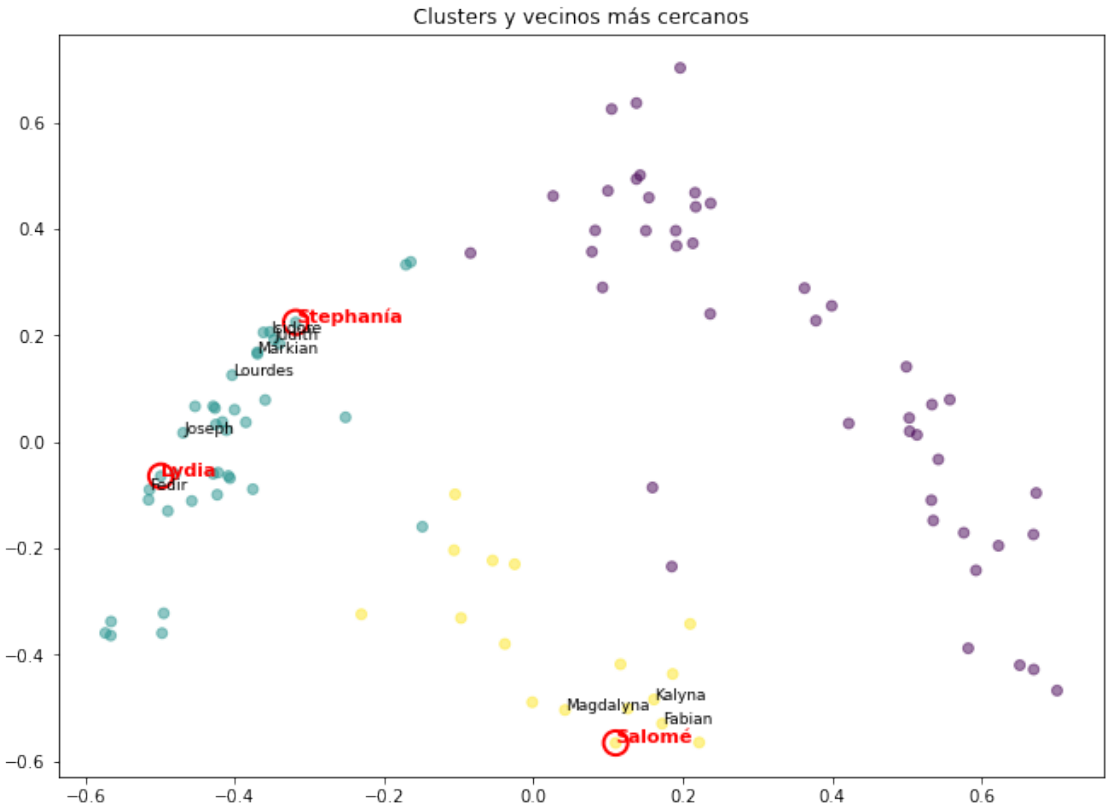
Stephanía (fila 89, cluster 1)
  -> Isidore  (distancia = 0.114)
  -> Markian  (distancia = 0.123)
  -> Judith  (distancia = 0.135)

Lydia (fila 54, cluster 1)
  -> Fedir  (distancia = 0.056)
  -> Joseph  (distancia = 0.125)
  -> Lourdes  (distancia = 0.253)
```

[76]:
```
#FINALMENTE, GRAFICAMOS E IDENTIFICAMOS A LAS CLIENTAS A LAS QUE DAREMOS UNA␣
 ↪RECOMENDACIÓN
#Y TAMBIÉN A LOS VECIMOS MÁS CERCANOS DE CADA UNA DE ELLAS
plt.figure(figsize=(11, 8))
plt.scatter(campos[:,0], campos[:,1], c=grupos, cmap='viridis', alpha=0.5)

for nombre, (i, cercanos) in vecinos_dict.items():
    plt.scatter(campos[i,0], campos[i,1], s=200, edgecolor='red',␣
 ↪facecolor='none', linewidth=2)
    plt.annotate(nombre, campos[i], fontsize=11, fontweight='bold', color='red')
    for j in cercanos:
        plt.annotate(resultado.loc[j,'Cliente'], campos[j], fontsize=9)
plt.title("Clusters y vecinos más cercanos")
plt.show()

print("Recomendación para Salomé: A Salomé se le recomendarán los mismos␣
 ↪productos que compró Magdalyna, Fabian y Kalyna.")
print("Recomendación para Stephanía: A Stephanía se le recomendarán los mismos␣
 ↪productos que compró Isidore, Markian y Judth.")
print("Recomendación para Lydia: A Lydia se le recomendarán los mismos␣
 ↪productos que compró Fedir, Joseph y Lourdes.")
print("Aunque en el gráfico 2d pudiera parecer que hay otros puntos más␣
 ↪cercanos a nuestros objetivos, se calcularon como los vecinos mas cercanos␣
 ↪en distancia dentro de los clusters. Se ve así por la reducción de␣
 ↪dimensiones de PCA.")
```



Recomendación para Salomé: A Salomé se le recomendarán los mismos productos que compró Magdalyna, Fabian y Kalyna.
Recomendación para Stephanía: A Stephanía se le recomendarán los mismos productos que compró Isidore, Markian y Judth.
Recomendación para Lydia: A Lydia se le recomendarán los mismos productos que compró Fedir, Joseph y Lourdes.
Aunque en el gráfico 2d pudiera parecer que hay otros puntos más cercanos a nuestros objetivos, se calcularon como los vecinos mas cercanos en distancia dentro de los clusters. Se ve así por la reducción de dimensiones de PCA